Internet of Things (IoT)

Materials & Notes

Module 1: Introduction to IoT

1.1 What is IoT?

The Internet of Things (IoT) refers to a network of physical devices that can collect, exchange, and process data using sensors, software, and communication technologies. These connected devices can communicate with other devices or systems through the internet or local networks. Examples include smart watches, smart home devices, connected vehicles, industrial sensors, and healthcare monitoring systems.

1.2 Characteristics of IoT

- Connectivity between devices and networks
- Sensing and collection of real-world data
- Data processing and analysis
- Automation and remote monitoring
- Scalability and device management

1.3 Applications of IoT

IoT is used in smart homes, healthcare, agriculture, transportation, manufacturing, energy management, retail, and smart cities. For example, smart agriculture systems can monitor soil moisture and environmental conditions to support better irrigation decisions.

Module 2: IoT Architecture and Devices

2.1 Basic IoT Architecture

A basic IoT system can be understood through sensing, network, processing, and application stages. Sensors collect information from the physical environment. Communication technologies transfer the information. Processing systems store and analyze the data, while applications present useful results to users.

2.2 Sensors and Actuators

A sensor detects a physical condition such as temperature, humidity, light, motion, pressure, or distance and converts it into usable data. An actuator performs an action based on a control signal, such as switching a motor, light, valve, or relay.

2.3 Microcontrollers

Microcontrollers are compact computing devices used to read sensor inputs and control outputs. Arduino boards are commonly used for beginner IoT projects, while Raspberry Pi can support more advanced applications and networked processing.

Module 3: Sensors and Data Collection

3.1 Common Sensors

- Temperature sensor – measures temperature
- Humidity sensor – measures moisture in air
- Light sensor – detects light intensity
- Motion sensor – detects movement
- Distance sensor – measures distance to an object

3.2 Sensor Data Collection

A sensor produces a measurement that can be read by a microcontroller. The controller can process the value, display it locally, or transmit it to another system. Repeated readings can be stored and analyzed to identify changes and patterns.

Module 4: IoT Communication and Protocols

4.1 Communication Technologies

IoT devices may use Wi-Fi, Bluetooth, Ethernet, cellular networks, or other communication technologies depending on distance, power, bandwidth, and application requirements.

4.2 MQTT

MQTT is a lightweight messaging protocol widely used in IoT. It follows a publish-subscribe model. A publisher sends messages to a topic, a subscriber receives messages from a topic, and a broker manages the communication between them. Its lightweight design makes it suitable for many connected-device environments.

4.3 HTTP

HTTP is another protocol that can be used when IoT devices communicate with web servers or APIs. For example, a device can send sensor readings to a server using an HTTP request.

Module 5: Cloud and IoT Data

5.1 IoT and Cloud Computing

Cloud platforms provide storage, computing, databases, analytics, and visualization services for IoT applications. Sensor data can be transmitted to a cloud service, stored, analyzed, and displayed through dashboards.

5.2 IoT Data Flow

A typical flow is: **Sensor** → **Microcontroller/Device** → **Network** → **Cloud/Server** → **Database** → **Dashboard/Application**. This flow allows users to monitor devices and make decisions using collected data.

Module 6: IoT Applications and Security

6.1 Real-World Applications

- Smart home – lighting, temperature, and security monitoring
- Healthcare – remote monitoring of health-related measurements
- Agriculture – soil and environmental monitoring
- Smart cities – traffic, waste, energy, and environmental monitoring
- Industry – equipment monitoring and automation

6.2 IoT Security

Security is important because IoT devices are connected to networks and may collect sensitive information. Basic security practices include strong authentication, access control, secure communication, software updates, device monitoring, and data protection.

Common Security Risks

- Weak passwords and authentication
- Unauthorized access
- Unsecured network communication
- Outdated device software
- Improper protection of stored data

Module 7: Mini Project – Smart Temperature Monitoring

A simple IoT temperature monitoring system can use a temperature sensor connected to a microcontroller. The sensor periodically collects temperature readings. The device can process the readings and send them to a server or cloud platform. The data can then be stored and displayed on a dashboard for monitoring.

Learning Outcomes

- Explain the fundamentals of IoT
- Identify major IoT components
- Understand sensors and communication protocols
- Explain how IoT data moves to cloud systems
- Identify common IoT applications and security practices
- Design a basic IoT monitoring solution